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Assignment Set – 1

Q1. What is Business Analytics? Discuss the role of Business Analytics in various domains of business. [5+5 = 10 Marks]

Business analytics is about using data and numbers to make business decisions. It has three parts: looking at what happened trying to predict what will happen and deciding what to do about it. Most companies only look at what happened. Some are getting better at predicting what will happen and making plans for it.

In marketing analytics helps people make decisions based on facts of just guessing. For example customer segmentation models group customers by what they buy so companies can make ads that're just right for each group. They also do something called A/B testing, where they try two things to see which one works better. And they use something called attribution modelling to see which ads actually worked so they can put their money into the ones that are doing well.

The finance people were some of the first to use analytics. They have models that try to figure out who might not pay back a loan. They also have models that look for patterns in the market to help them make trades.. They have systems that can spot fake transactions, which helps prevent fraud. If they did not have these models they would not be able to keep up with all the transactions that happen every day.

Companies that make and sell things use analytics to figure out how much of something they will need so they can. Ship the right amount. For example Walmart uses analytics to make sure they have the right amount of stuff in their stores so they do not run out. Amazon does something they use models to figure out where people will want things and they put those things in warehouses near those places before people even order them.

In healthcare analytics is used to try to predict if a patient will have to come to the hospital and to make sure there are enough nurses on duty. Human resources people use analytics to try to figure out which employees might quit so they can try to keep them.. Companies that make things use analytics to keep their machines running by spotting problems before they happen. The main idea is the same use data and numbers to make good decisions instead of just guessing. Business analytics is, about using data and numbers to make business decisions and it is used in many different areas, including marketing, finance, supply chain management, healthcare and manufacturing to replace guesswork with evidence and business analytics is helping companies make better decisions every day.

Q2(A). Discuss any one purpose of using Conjoint Analysis in detail. [5 Marks]

Conjoint analysis is a technique that market researchers use to figure out what consumers like about a product or service. It is really useful when designing products because it helps teams make decisions about what features to include.

Consider a company that makes smartphones. They have to choose between a better camera, a bigger battery, a thinner phone and a lower price. They cannot do all of these things at the time because it would be too expensive and hard to make. So they do a conjoint study where they show people versions of the phone and ask them which one they like best. For example one phone might have a camera and a big battery but cost a lot of money. Another phone might have a camera and a really big battery but cost less money.

When they look at the data from the study they can see what people like about each version of the phone. If people really like phones with batteries but they do not care as much about the camera then the company knows what to focus on. This way the company can make a phone that people will really like and want to buy.

Lots of companies, like car makers and technology companies use conjoint analysis to design their products. It also helps them figure out how much to charge for their products. By using conjoint analysis companies can see what will happen if they add a feature or change the price of a product. This helps them make decisions and not waste money on things that people do not want.

Q2(B). Explain the following terms and how they are calculated: TRP, GRP. [5 Marks]

TRP or Target Rating Point measures how many people in a group see an ad. For example if the target group is women aged 25 to 40 and 20 percent of them see the ad 4 times then TRP is 80. You can also calculate it as (number of people in target group who see the ad / number of people in target group) x 100.

Media planners use TRP to see if an ad is reaching the people.

* GRP or Gross Rating Point uses the way of calculating but looks at the total population, not just a specific group.

* If 35 percent of all TV viewers see an ad 3 times then GRP is 105.

* A high GRP means many people see the ad. It does not say if they are the people the ad is meant for.

This difference is important because an ad can have a GRP but low TRP.

This means most people who see the ad are not the ones the ad is meant for which's a waste of money.

Media buyers use TRP to compare how different ad placements, like TV or online ads reach the target audience.

This helps them spend their budget better.

Q3. Discuss MDS and Perceptual Map in detail. [5+5 = 10 Marks]

Multidimensional Scaling or MDS is a way to take a bunch of information about how similar or different things are and turn it into a map. On this map things that are really similar are close together. Things that are really different are far apart. For example if people think Brand A and Brand B are very similar MDS will put them close on the map. If people think Brand A and Brand B are very different they will be apart on the map.

There are two kinds of MDS. One is called Metric MDS. It tries to keep the distances between things the same. This kind of MDS is good when we have measurements. The other kind is called non-metric MDS. It only looks at which things are more similar or less similar than others. This kind of MDS is good when we have information from surveys where people just say which things are more similar or less similar.

We can tell how good the map is by looking at something called the stress statistic. If the stress is low the map is a representation of how similar or different things are.

The map that MDS makes is called a map. It is a way to see where each brand or product is in relation to others. We can use it to understand how people think about brands or products. The axes on the map are like lines that we can use to understand what makes each brand or product unique. For example one axis might be about whether something's expensive or cheap and the other axis might be about whether something is traditional or new.

Marketers like maps because they give a quick view of how brands compete with each other. If two brands are close together on the map they compete with each other a lot. They probably compete on price. If there is a space on the map it might mean that there is a group of people who are not being served by any brand and that could be an opportunity for a new brand or product. For example in the fast food industry a perceptual map might show that there are a lot of brands that're similar and compete on price but there might be a space for a brand that is both affordable and healthy.

Multidimensional Scaling or MDS helps us make these maps. Perceptual maps are useful for understanding how people think about brands or products. They help marketers understand

how to compete with brands and how to find new opportunities. Multidimensional Scaling or MDS is a tool, for making perceptual maps.

Assignment Set – 2

Q4. What is Propensity Modelling? Briefly discuss different types of Propensity Modelling. [5+5 = 10 Marks]

Propensity modelling uses information and supervised learning to figure out how likely a person is to do something specific like buying a product stopping a subscription answering an offer or clicking on an ad. Of treating all customers the same way each person gets a score that shows how likely they are to do something, which helps companies make targeted and cost-effective moves.

Propensity to purchase models give scores to customers based on how they are to buy something. A company that sells things online might use a kind of model that looks at things like what a person has browsed how long it has been since they last bought something what they have put in their cart and whether they open emails. People who get scores are sent special offers and people who get low scores are not sent as much which helps reduce waste and makes the companys money go further.

Propensity to churn models try to predict which customers are likely to leave. Companies like telecom operators, streaming services and software platforms really need these models because it usually costs a lot more to get a customer than to keep an existing one. If a customer is thought to be at risk of leaving they might get a special loyalty offer, a personal check-in or a contract incentive before they decide to switch.

Propensity to engage models try to figure out who will interact with a marketing message, like opening an email clicking on a notification or watching an ad. This helps companies choose the time and way to send their messages. Propensity to convert models focus on turning people who're interested into paying customers by giving them scores based on how likely they are to buy something.. Propensity to respond models, which are used a lot in direct marketing try to predict who will act on a specific offer. The basic way these models work is the same using things, like regression, random forest, XGBoost or neural networks to look at past results but what is different is what the model is trying to predict.

Q5. Briefly discuss any five Digital Marketing techniques. [10 Marks]

Search Engine Optimisation or SEO for short is about making a website show up more often in search results. This means doing some research on keywords making sure the website is easy to use and looks good and getting websites to link back to it. The good thing about SEO is that it can bring people to the website without having to pay for each click. However it takes a time to see results and it needs to be constantly updated because search engines are always changing.

Pay-Per-Click or PPC is a way to advertise on websites like Google and social media. When someone clicks on an ad the person who made the ad has to pay. To make PPC work well you have to keep trying versions of the ad and make sure the page it links to is good. It can bring in people quickly. It stops working as soon as you stop paying. So it is an idea to use it along with SEO.

Social Media Marketing is when you use platforms like Instagram, LinkedIn, YouTube and TikTok to reach people. Each platform is different. Has its own type of audience and content. For example LinkedIn is good for stuff while TikTok is better for short fun videos. You can also pay to advertise on these platforms and choose who you want to reach.

Content Marketing is when you make and share things like blog posts, videos and podcasts to attract people. If a company that does cybersecurity makes a guide on how to respond to incidents it can make people trust them. Then when someone is ready to buy something they are more likely to choose that company. Content Marketing and SEO work well together because good content can bring in both people and links from websites.

Email Marketing is still one of the ways to reach people online. It works best when you send emails to groups of people and use automation to make it easier. For example you can send reminders to people who left something in their cart or recommend products based on what they like. When you do it this way many more people will open your emails.. When you have an email list you own it so you do not have to worry about changes, to social media algorithms that can make it harder to reach people.

Q6. Elaborate RPA. Discuss the 5-step implementation approach of RPA. [5+5 = 10 Marks]

Robotic Process Automation uses software bots to do tasks on computers that humans normally do. These tasks are usually repetitive. Follow certain rules.

The bots work by interacting with the computer screen like clicking buttons typing text and moving between menus. They can do this without needing to connect to the computers backend systems.

This makes it easy to use RPA with computer systems where changing the code is not possible. Some popular RPA platforms are UiPath, Automation Anywhere and Blue Prism.

The first step is to identify and assess tasks to see if they are suitable for automation.

- * Tasks that are good for automation are those that happen often follow rules and have digital inputs.

- * Tasks that need a lot of judgement or have many exceptions are not good for automation.

A scoring system helps decide which tasks to automate first based on how they can be automated and how much they will help the business.

The second step is to design the process.

Every step the bot will take, like clicking, typing and making decisions is written down in a document.

This is important because the bot can't make decisions on its own. If something isn't written down it will fail.

The third step is to build the bot.

Using a tool developers create the automated workflow by adding steps and configuring each one.

Most RPA tools are easy to use. Don't require a lot of coding but some complex tasks might need some coding.

The fourth step is testing.

The bot is tested with different scenarios, in a safe environment to make sure it works correctly.

The people who will use the bot also test it to make sure it produces the results.

The fifth step is to deploy and monitor the bot.

The bot is put into production. Run as needed.

The performance of the bot is. Maintenance is done regularly to make sure it keeps working.

Many companies have a team to manage their bots measure how well they are working and find new opportunities to automate tasks.